A hybrid data platform for monitoring, analyzing and predicting urban activity in NYC

City Live Monitor

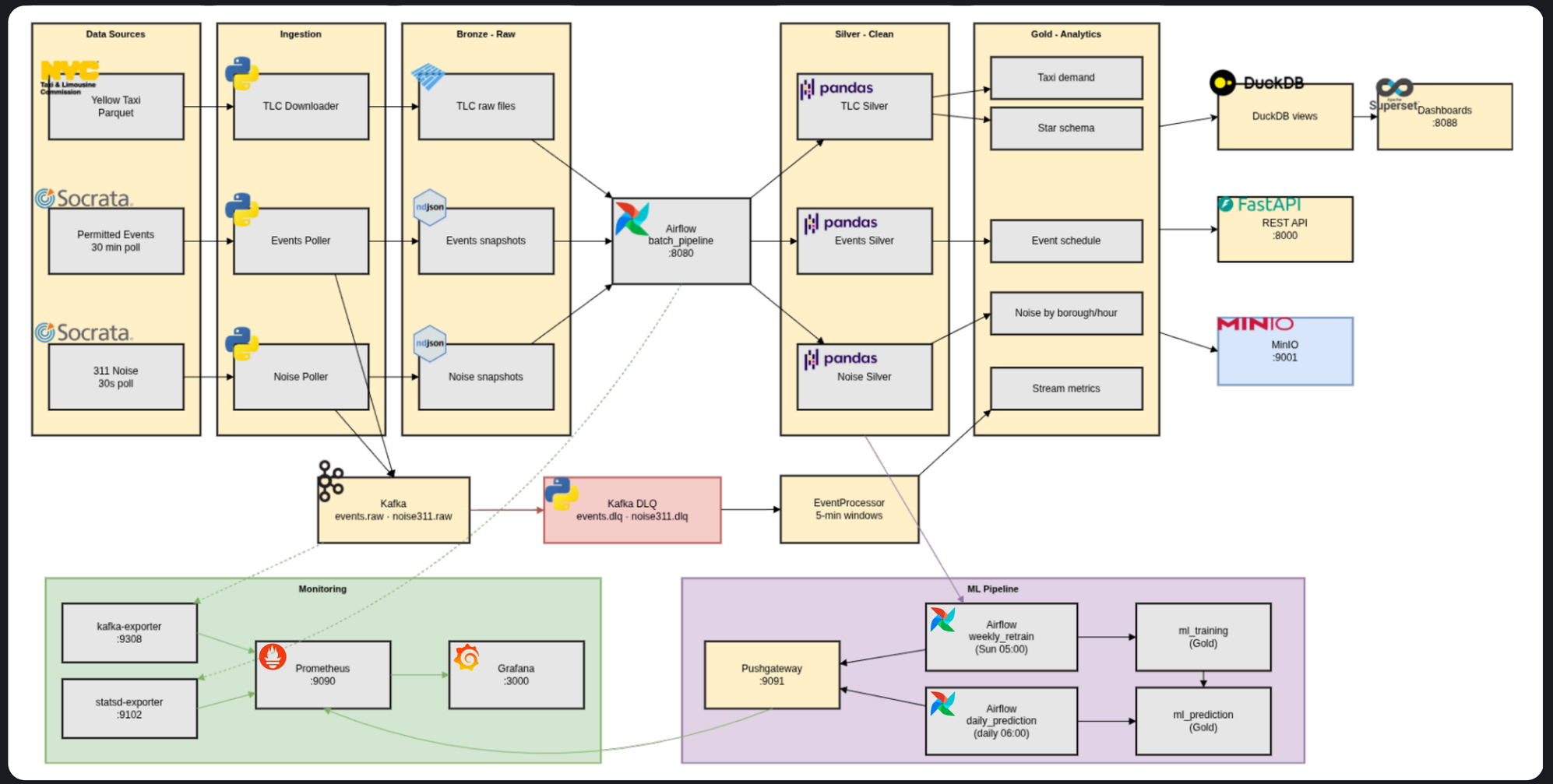
The Path to Operational Maturity

STATUS: PRODUCTION-READY

Gabriel Silva (113786) | Hugo Castro (113889)

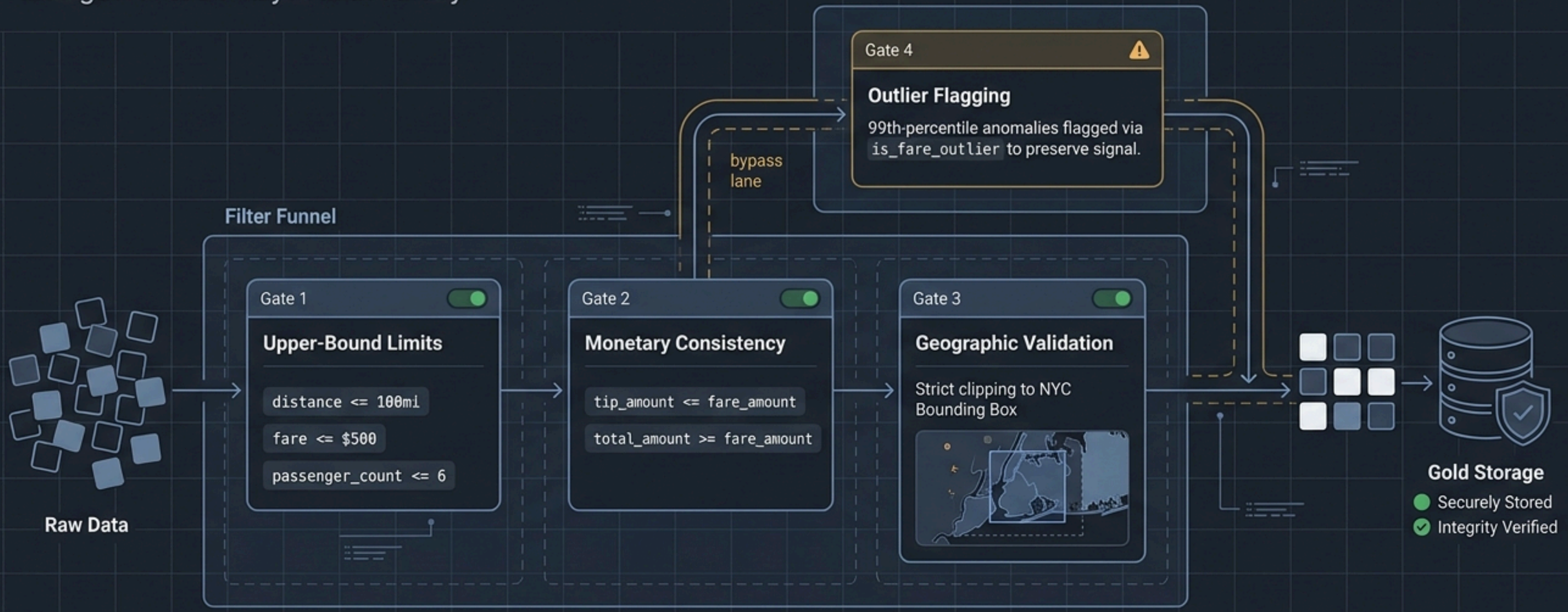
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Medallion Architecture



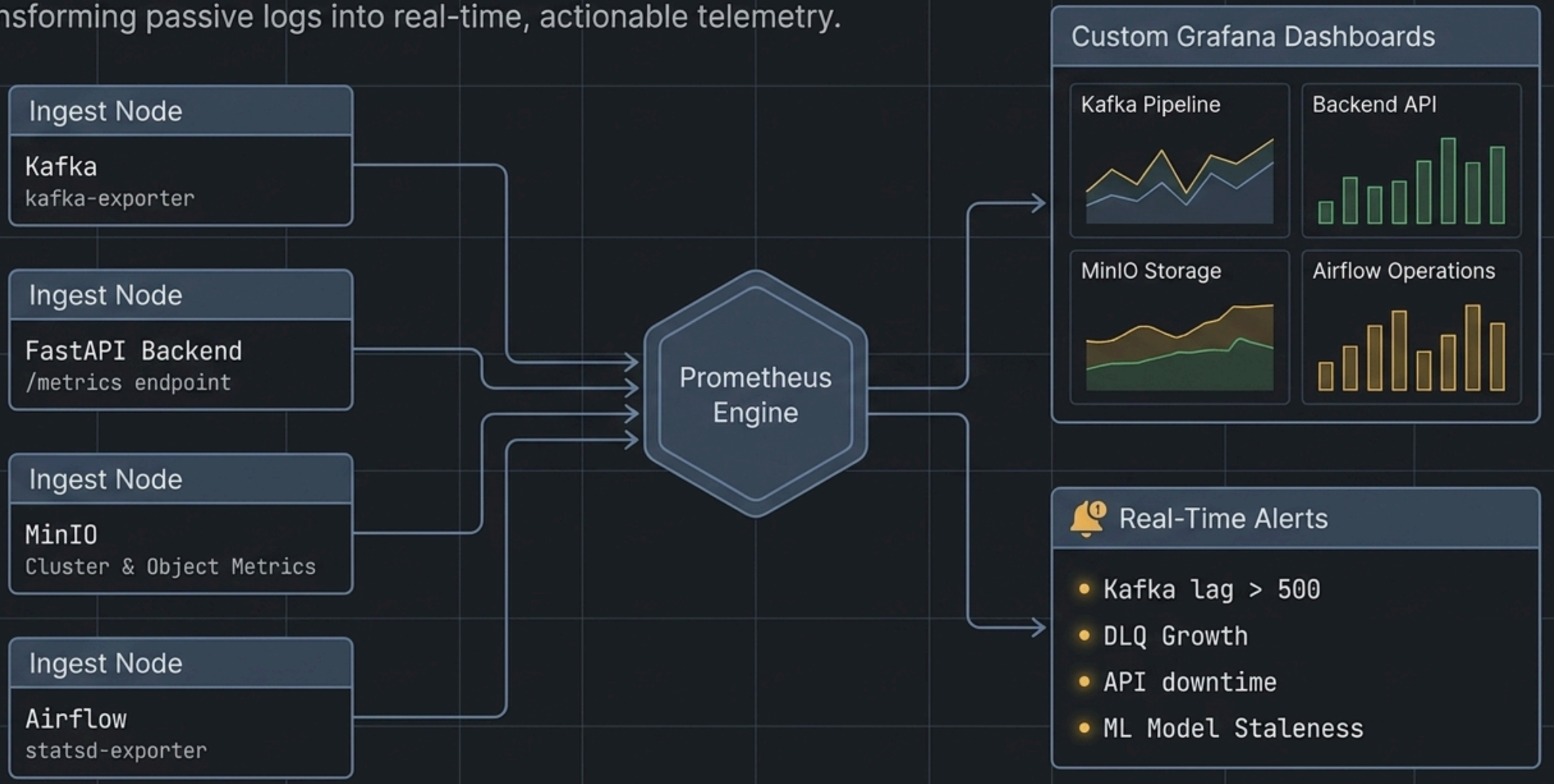
Ironclad Data Integrity

Advanced upper-bound filtering and consistency rules guarantee analytical accuracy.



Total Platform Visibility

Transforming passive logs into real-time, actionable telemetry.



ML Logic & Implementation Summary

Key components and operational workflow of the machine learning pipeline.

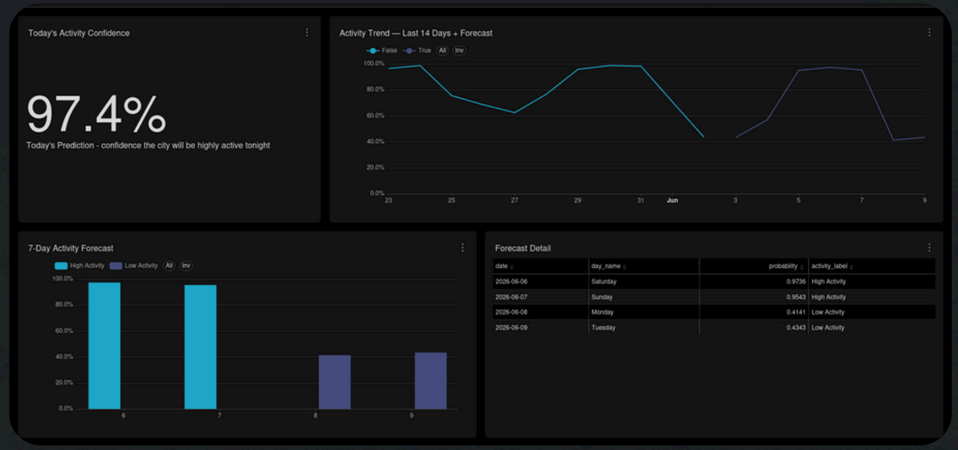
ML Model & Strategy

- **Model:** HistGradientBoostingClassifier.
- **Objective:** Predict high activity probability.
- **Training:** Weekly retrain (Sundays) on historical data.



Operational Workflow

- **Prediction:** Daily batch run at 06:00 UTC.
- **Monitoring:** Metrics (ROC-AUC, F1) pushed to Pushgateway.



Results

124.2M
raw records
Bronze ingestion

103.1M
cleaned records
Silver layer

10
Gold artefacts
analytics + API + ML

11
test files
automated validation

Data Products

- **102.9M -> 38.9K** taxi trips into demand groups
- **71.6K** unique permitted events
- **100.1K** noise complaints processed
- **26.1K** Kafka 5-minute metric windows

Serving + Operations

5	Superset dashboards
4	Grafana dashboards
3	Airflow DAGs
API	Gold + metrics

ML Activity Classifier

84.5% accuracy

91.6% ROC-AUC

7-day forecast

Live Demo

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